

Uttkarsh Thakur

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SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

AI/ML: RAG Pipelines, Semantic Search, Vector Databases, FAISS, AST Parsing, NLP, Transformers, LLMs, OpenAI API

ML Libraries: LangChain, Sentence-Transformers, TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas

Backend & Web: FastAPI, Flask, Spring Boot, Spring Data JPA, REST API Design, React.js, Node.js, HTML/CSS

Databases: PostgreSQL, MySQL, MongoDB, H2, Flyway Migrations

Developer Tools: Git & GitHub, Docker, Maven, JUnit 5, Jupyter Notebook, Google Colab, VS Code

PROJECTS

CodeNav AI | *FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython* 2026

- Built an **AI-powered search tool** that lets a developer ask questions about an unfamiliar codebase in plain English and get back the relevant code, using **large language models** and **semantic search (RAG)**.
- Improved answer quality by splitting code along real function and class boundaries instead of fixed-size blocks, so results come back complete and in context.
- Cut the time to open a codebase from **over 20 seconds to under 0.1 seconds**, a **99.5% improvement**, by saving the search index to disk instead of rebuilding it each time.
- Delivered the tool as a web application with a **React.js** interface and a **FastAPI** backend.

BillSync | *Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker* 2026

- Built a **full-stack web application** for splitting group expenses: members log what they paid, and the app works out who owes whom, backed by a **Java Spring Boot** REST API of **16 endpoints** and a **React** and **TypeScript** interface.
- Solved a rounding problem so every split adds up to the **exact total** with no lost fractions, and developed a **settle-up feature** that turns a tangle of debts into the fewest payments possible.
- Designed the **PostgreSQL** database with version-controlled migrations and wrote **73 automated tests**; tuned a slow screen that made **22 database calls** down to **1**.

Motion Sensing System | *TensorFlow/Keras, VGG16, Flask, Streamlit, OpenCV, Python* 2024

- Built a system that **recognises human activities from live video**, adapting a pre-trained image-recognition model (**VGG16**) rather than training one from scratch.
- Delivered it as a live dashboard that displays activity predictions as they happen.

RESEARCH & TALKS

Agreement Loops in Multi-Agent Systems | *Seminar Talk, Literature Review* 2026

- Reviewed current research on **how independent AI agents reach agreement** with one another across the wider field of distributed AI.
- Presented the findings as a **seminar talk**, covering what works reliably today and which problems remain unsolved.

EDUCATION

Bennett University Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering | **CGPA: 8.17/10**

Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

DAV Public School Graduated 2022

Senior Secondary Education (Class XII)

Ranchi, India

CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera Nov. 2024

ACHIEVEMENTS

Ranked **101st nationally** at **Smart India Hackathon 2024**

CodeChef: 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)